AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 10 · STUDENT LESSON

Evaluation pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Keep the plan. Preserve every outcome. Name the denominator.

Build a versioned evaluation record that reports completion coverage alongside pass rate and preserves missing/skip distinctions.

Keep one habit: Manifest → complete six-case ledger → paired-picture comparison → record-integrity probes → executable summarizer → independent eight-case counts.

MAIN LESSON · SECTION 1

Keep every planned case visible when computing a result

> Pass rate and completion coverage need different denominators.

Learning goals

I can explain the **mechanism** and terminology of evaluation pipelines.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve fixture and repeated-case evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Integration tests and fixtures. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My starting explanation

End goal

Explain evaluation pipelines using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of evaluation pipelines

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Version the plan and declare how each status is counted

> A summary cannot repair absent or selectively removed raw outcomes.

Begin with six planned checks

Suppose a class plans six service checks. Five finish, with four passes and one failure; the sixth is explicitly skipped. “Four passed” is incomplete without saying what happened to the rest. An evaluation pipeline keeps the plan, versioned cases, procedure, raw outputs and scoring rule together before it reports a percentage.

A missing result record is an error in that record set; an explicit skipped record is an unassessed case, not a success.

Define evaluation pipelines and rerun versions

Evaluation pipelines preserve case versions, procedure versions and **raw outcome**s before producing metrics. A rerun must identify what changed.

Complete manifest and denominator policy

Pipeline item	Declared local exercise
Case suite	ROOM-CAPACITY, case version classroom-v1
Frozen procedure	fixed-lookup-stub-v1; no actual model call
Planned IDs	C1–C6, one explicit outcome record per ID

Pipeline item	Declared local exercise
Metric	Exact nonnegative integer capacity agreement with reference
Completed-case pass rate	Passed completed cases / all completed cases
Coverage	Completed cases / all planned cases
Skipped case	Keep ID and reason; exclude from completed pass denominator but retain in coverage
Missing/duplicate/unknown ID	Reject inconsistent record set; do not silently classify
No completed cases	Coverage zero; completed-case pass rate unavailable

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Compute the counts from the case-level ledger

> A skipped row stays in the plan and must not appear as a passed completed case.

Original pipeline stages and missing-result requirement

Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Complete six-case input/reference/output ledger

Planned ID / room input · Reference capacity · Local stub output · Explicit status · Outcome

> **Planned ID / room input:** C1 / A
> **Reference capacity:** 12
> **Local stub output:** 12
> **Explicit status:** completed
> **Outcome:** pass

> **Planned ID / room input:** C2 / B
> **Reference capacity:** 8
> **Local stub output:** 8
> **Explicit status:** completed
> **Outcome:** pass

> **Planned ID / room input:** C3 / C
> **Reference capacity:** 10
> **Local stub output:** 10
> **Explicit status:** completed
> **Outcome:** pass

> **Planned ID / room input:** C4 / D
>
> **Reference capacity:** 6
>
> **Local stub output:** 6
>
> **Explicit status:** completed
>
> **Outcome:** pass

> **Planned ID / room input:** C5 / E
>
> **Reference capacity:** 4
>
> **Local stub output:** 5
>
> **Explicit status:** completed
>
> **Outcome:** fail

> **Planned ID / room input:** C6 / F
>
> **Reference capacity:** 2
>
> **Local stub output:** not run
>
> **Explicit status:** skipped: declared classroom skip
>
> **Outcome:** unassessed

> **Planned ID / room input:** Totals
>
> **Reference capacity:** 6 planned
>
> **Local stub output:** 5 completed
>
> **Explicit status:** 4 pass / 1 fail / 1 skip
>
> **Outcome:** Pass4/5=80%; coverage 5/6≈83.33%

State the denominator before calculating

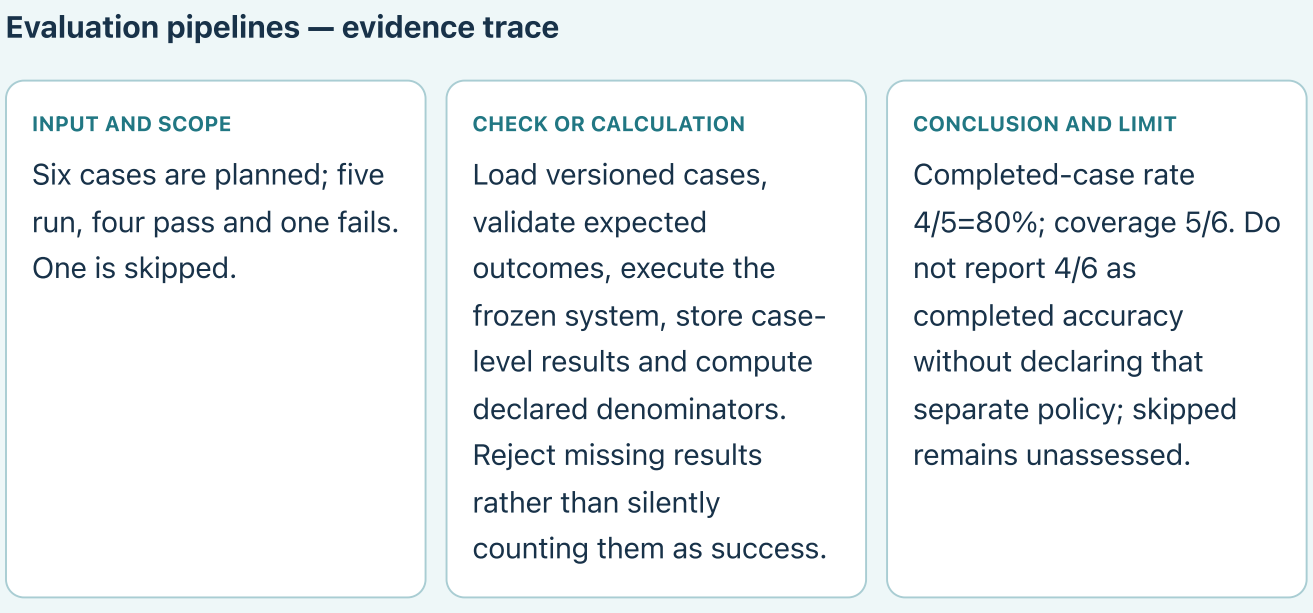
Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change the record set and predict the denominator effect

> Keep the original plan and raw outcomes rather than overwriting them.

Current task and response

- Six cases are planned; five run, four pass and one fails. One is skipped. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict deleting the shared pictured failure

- Change: Remove the shared failed third case after scoring.

▼ **Check after predicting**

Prediction: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading.

Reason: The population changed after seeing failures, concealing a weakness.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare shared cases without deleting inconvenient results

> A post-result change in the evaluated population needs an explicit label.

Main planned/completed counts and bounded rates

Six cases are planned; five run, four pass and one fails. One is skipped.

Completed-case rate 4/5=**80%**; coverage 5/6. Do not report 4/6 as completed accuracy without declaring that separate policy; skipped remains unassessed.

Keep the reasoning's scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate picture flags, references and denominator

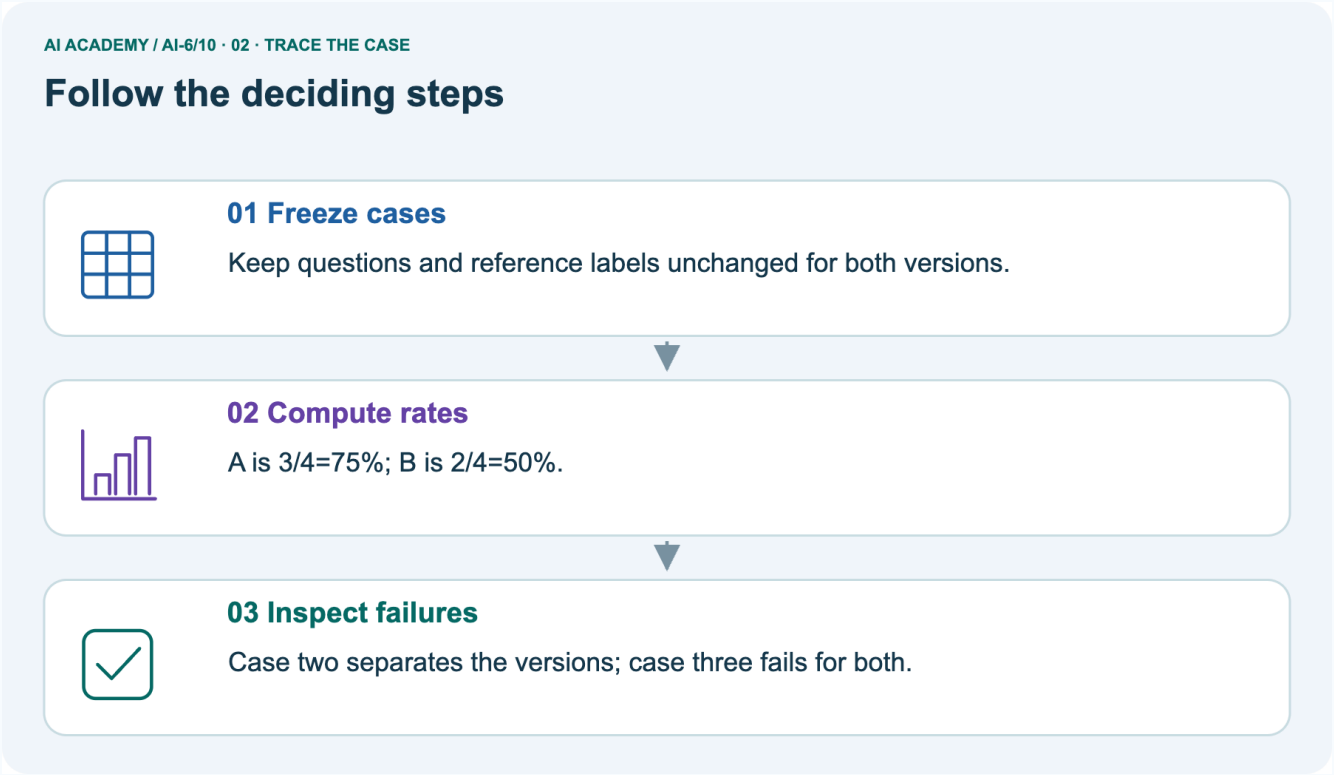
Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

A scores [1,1,0,1]; B scores [1,0,0,1]. Each 1 means agreement with the reference answer.

The main six-case completion example and the four-case picture are separate. Picture A has supplied correctness flags [1, 1, 0, 1], B [1, 0, 0, 1]; 1 means agreement with a reference, not the predicted class. Full rates are 3/4=75% and 2/4=50%. Case2 separates the versions; case 3 fails for both.

Removing case 3 after seeing the failures changes the rates to3/3 and 2/3, but conceals a known weakness if it replaces the original report. A useful declared subgroup can be reported separately with its selection rule and full counts retained.

Trace the full paired outcomes



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze cases	Keep questions and reference labels unchanged for both versions.
2. Compute rates	A is 3/4=75%; B is 2/4=50%.
3. Inspect failures	Case two separates the versions; case three fails for both.

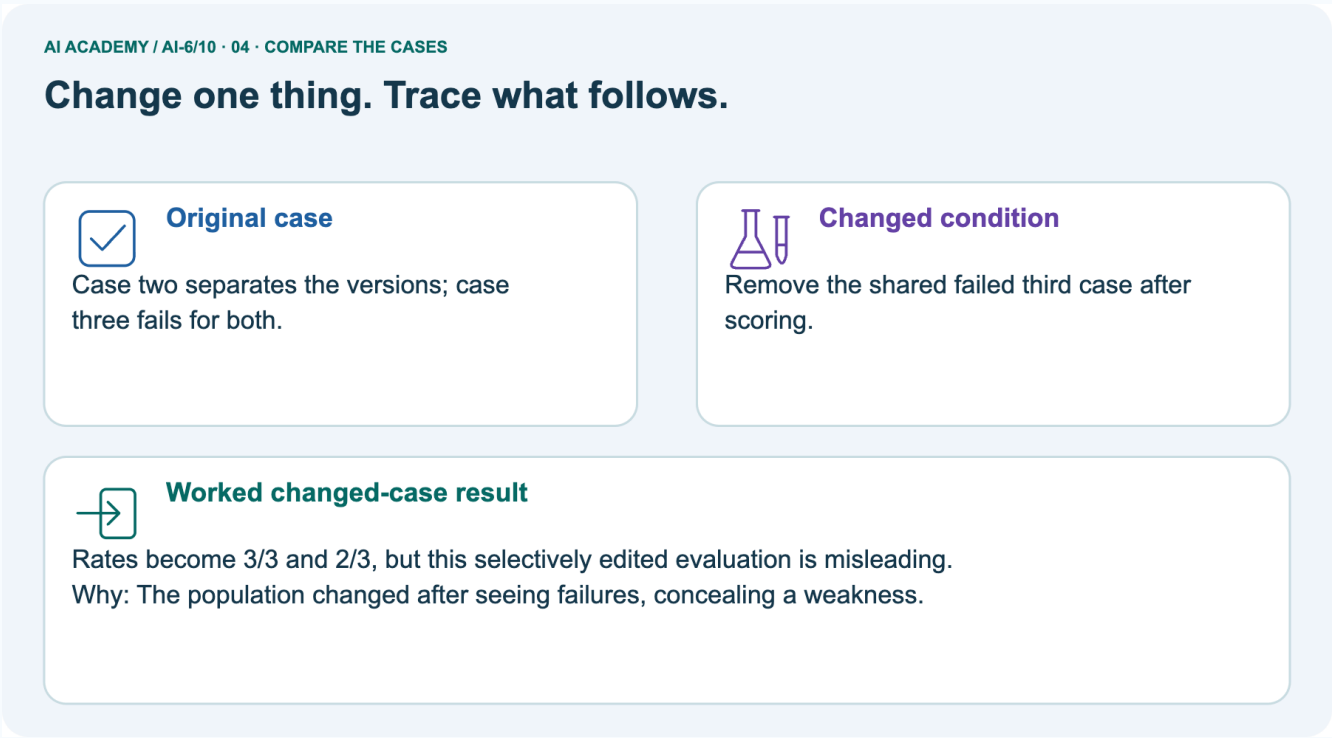
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A pipeline preserves comparable cases, version labels and scoring. Report the denominator and disagreements alongside percentages.

Predict removing the shared failure

- Change one thing: Remove the shared failed third case after scoring. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain shared/disagreeing cases and limited evidence

- Four fictional cases illustrate a procedure, not broad quality or a stable future success rate.
- Read the labelled picture: Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

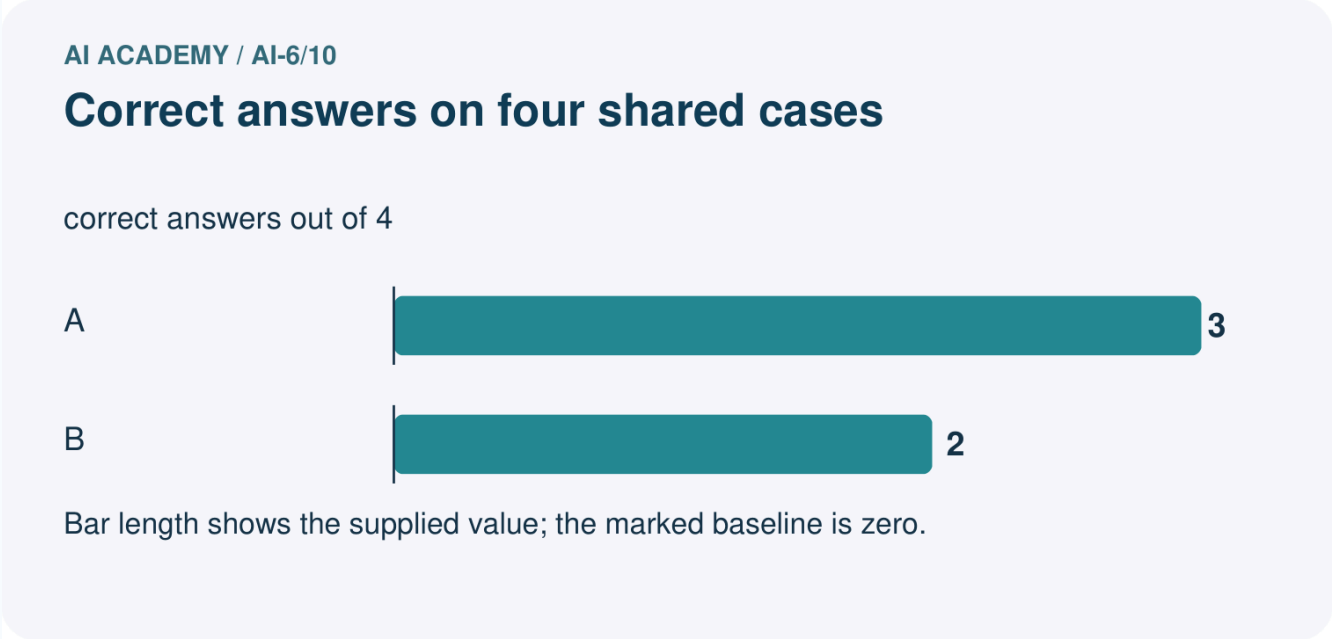
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

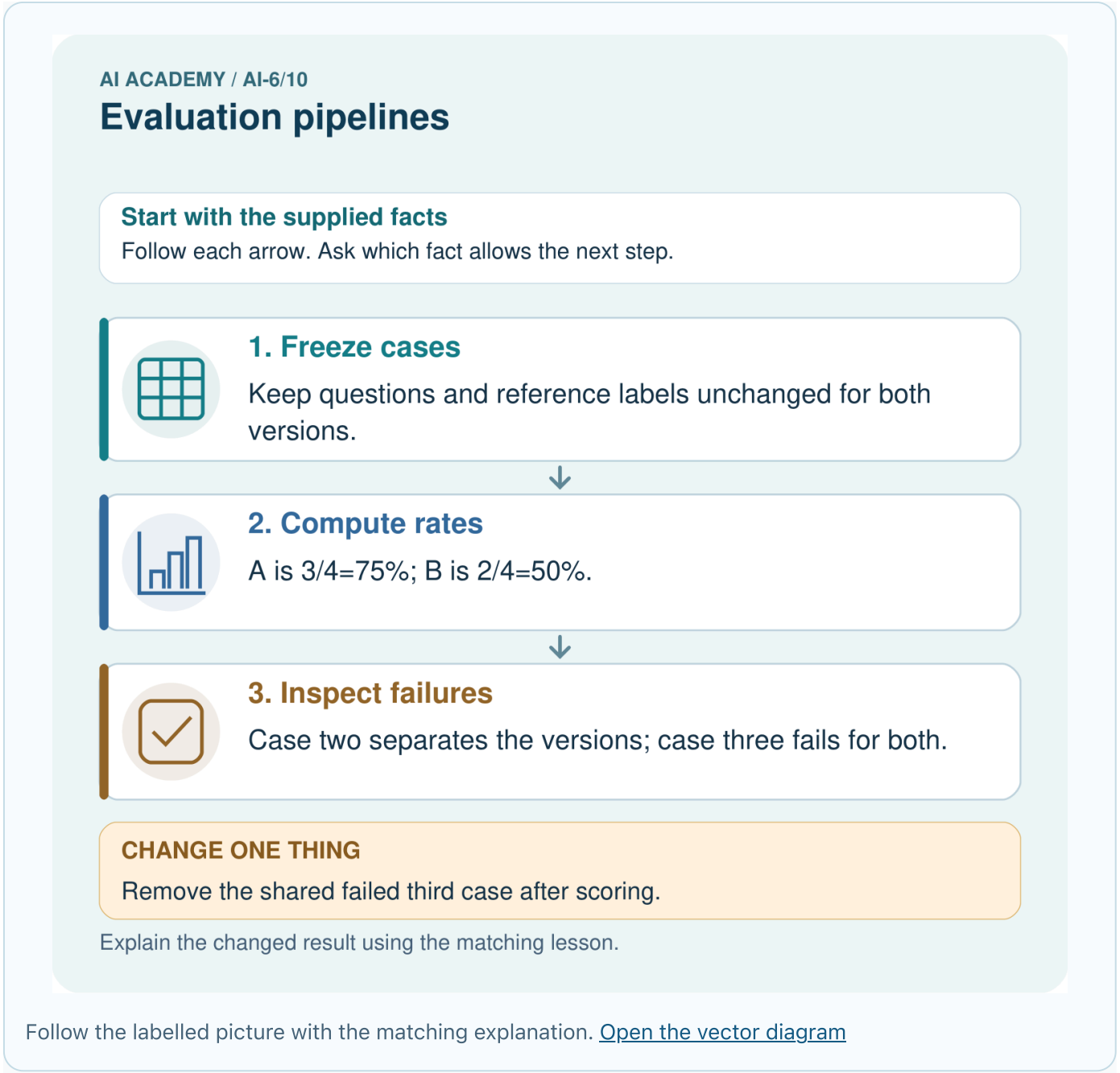
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Evaluation pipelines

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit the record set before trusting its percentage

> Completion and pass rates must not hide one another.

Core missing-evaluation qualification

Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete missing, duplicate, wrong-output and no-completion cases

Record or metric probe	Expected interpretation
Six planned, five completed, four passes	Completed pass 4/5; coverage 5/6; one explicit unassessed skip
Case C6 disappears from results	Reject missing record, not automatically success or intentional skip
Duplicate ID or unexpected result ID	Reject inconsistent result set
Completed actual output is wrong type or value	Completed contract failure; do not discard it from denominator
All planned cases explicitly skipped	Coverage0; completed pass rate undefined, not perfect
Only a pass count is reported	Require completion/skip counts and denominator before interpreting
Known failures removed after scoring	Preserve full original report and label the changed subset
Same deterministic stub replayed	Rechecks procedure; no independent cases or live reliability evidence added

Guided explanation and repair

- A report omits this qualification: Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise counts, denominators and versioned comparisons

Keep planned, completed and passed counts in separate boxes.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Evaluation pipelines”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual missing-evidence check

- Use the worked case for Evaluation pipelines. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Paired-picture response

- Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the independent eight-case counts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete versioned ledger and coverage audit

> Missing records must be visible errors; intentional skips must be explicit unassessed rows.

Current six-case evaluation investigation

Six cases are planned; five run, four pass and one fails. One is skipped. Show the intermediate steps and give the conclusion with its scope.

Complete runnable stub, record validation and scoring

```
"""Versioned local stub evaluation with explicit skipped rows and denominators."""
manifest={'suite':'ROOM-CAPACITY','case_version':'classroom-v1',
          'procedure_version':'fixed-lookup-stub-v1',
          'scoring':'exact nonnegative integer capacity equality',
          'skip_policy':'exclude from completed pass denominator; retain in planned coverage'}
cases=[{'id':f'C{i+1}','input':room,'expected':expected}
       for i,(room,expected) in enumerate(zip('ABCDEF',[12,8,10,6,4,2]))]
STUB={'A':12,'B':8,'C':10,'D':6,'E':5,'F':2}
def execute_fixture(cases):
    records=[]
    for case in cases:
        if case['id']=='C6':
            records.append({'id':'C6','status':'skipped','reason':'declared classroom skip'})
        else:
            records.append({'id':case['id'],'status':'completed','actual':STUB[case['input']]})
    return records

def summarize(cases,records):
    if not cases:raise ValueError('Need planned cases')
    if any(type(c) is not dict or set(c)!={'id','input','expected'}
           or type(c['id']) is not str or not c['id']
           or type(c['input']) is not str or not c['input'] for c in cases):
        raise ValueError('Invalid planned-case schema')
    for record in records:
        if (type(record) is not dict or record.get('status') not in ('completed','skipped')
            or type(record.get('id')) is not str or not record['id']):
            raise ValueError('Invalid result record')
        fields=({'id','status','actual'} if record['status']=='completed' else
{'id','status','reason'})
        if set(record)!=fields:raise ValueError('Unexpected or missing result fields')
    ids=[c['id'] for c in cases]
    if len(ids)!=len(set(ids)):raise ValueError('Duplicate planned ID')
    if any(type(c['expected']) is not int or c['expected']<0 for c in cases):
        raise ValueError('Invalid reference capacity')
    result_ids=[r['id'] for r in records]
    if len(result_ids)!=len(set(result_ids)) or set(result_ids)!=set(ids):
        raise ValueError('Need exactly one explicit record for every planned ID')
    by_id={r['id']:r for r in records}
    ledger=[]
    for case in cases:
        record=by_id[case['id']]
```

```
        if record['status']=='skipped':
            if type(record.get('reason')) is not str or not record['reason']:raise
ValueError('Skipped record needs a reason')
            ledger.append({**case,**record,'pass':None})
        elif record['status']=='completed':
            if 'actual' not in record:raise ValueError('Completed row missing actual output')
            actual=record['actual']
            passed=type(actual) is int and actual>=0 and actual==case['expected']
            ledger.append({**case,**record,'pass':passed})
        else:raise ValueError('Unknown result status')
    completed=sum(r['status']=='completed' for r in ledger)
    passes=sum(r['pass'] is True for r in ledger)
    return {'planned':len(cases),'completed':completed,'passed':passes,
            'failed':completed-passes,'skipped':len(cases)-completed,
            'coverage':completed/len(cases),
            'completed_pass_rate':passes/completed if completed else None,
            'ledger':ledger}

records=execute_fixture(cases)
summary=summarize(cases,records)
assert
(summary['planned'],summary['completed'],summary['passed'],summary['failed'],summary['skipped'])==
(6,5,4,1,1)
assert summary['completed_pass_rate']==.8 and summary['coverage']==5/6
assert records==execute_fixture(cases)  # same stub replay, not independent evidence
print('Manifest:',manifest)
for row in summary['ledger']:print('Raw case-level row:',row)
print('Completed pass rate:',summary['passed'],'/',summary['completed'])
print('Coverage:',summary['completed'],'/',summary['planned'])
try:summarize(cases,records[:-1])
except ValueError:print('Missing result record rejected; it is not silently a pass or a skip.')
else:raise AssertionError('Omitted planned case accepted')
all_skipped=[{'id':c['id'],'status':'skipped','reason':'not run'} for c in cases]
assert summarize(cases,all_skipped)['completed_pass_rate'] is None
# Separate supplied picture: correctness flags, not predicted class labels.
a=[1,1,0,1];b=[1,0,0,1]
assert sum(a)/4==.75 and sum(b)/4==.5
print('Separate full picture:',sum(a),'/4;',sum(b),'/4')
print('Selectively deleting known shared failure changes rates to 3/3 and 2/3; preserve the
full original report.')
print('Local stub only: no real model run, broader quality estimate or runtime measurement.')
```

Expected rows, guard conditions and metric scope

1. Predict the six raw rows and both rates before running or tracing. The fixed lookup stub intentionally gives the wrong value for E and explicitly skips C6; it is not a real model. The summarizer validates case/result fields, unique matching IDs, explicit skip reasons and recorded actual outputs before aggregation. It rejects an omitted C6 record and does not let result records overwrite reference fields.

   Wrong output types count as completed failures under the declared scoring rule. With all rows skipped, pass rate is None. The separate pictured four-case flags are not folded into the six-case denominator. Save manifest, raw rows, rates and actual run/trace status together.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compute both rates from fresh planned-case counts

> State what happened to the uncompleted cases before making a quality claim.

Complete independent eight-case summary task

Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the manifest, raw outcomes and unchanged original report

> Make rerun changes identifiable rather than silently replacing results.

Project task and artifact

Add a labelled evaluation pipelines **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of evaluation pipelines.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a percentage that exposes its denominator

> Retain skipped cases and failed cases alongside passes.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of evaluation pipelines

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Pipeline manifest	Versions and expected stages of an evaluation run
Coverage denominator	The planned cases eligible for evaluation
Raw outcome	The stored case-level result before aggregation

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the service reasoning and its limits**

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Evaluation pipelines — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Six cases are planned; five run, four pass and one fails. One is skipped.	Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.	Completed-case rate 4/5=80%; coverage 5/6. Do not report 4/6 as completed accuracy without declaring that separate policy; skipped remains unassessed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.

By the end, I can	How I will show it
Explain the mechanism and terminology of evaluation pipelines	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.
- Which labelled step matches this detail? Case two separates the versions; case three fails for both.
- 1. Freeze cases
- 2. Inspect failures
- 3. Compute rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect failures Case two separates the versions; case three fails for both. A pipeline preserves comparable cases, version labels and scoring.

Report the denominator and disagreements alongside percentages.

Changed-case question and optional hint

- Changed-case practice: Remove the shared failed third case after scoring. Which conclusion is supported by the supplied facts?
- 1. Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. The population changed after seeing failures, concealing a weakness.

Repair and retain the record

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

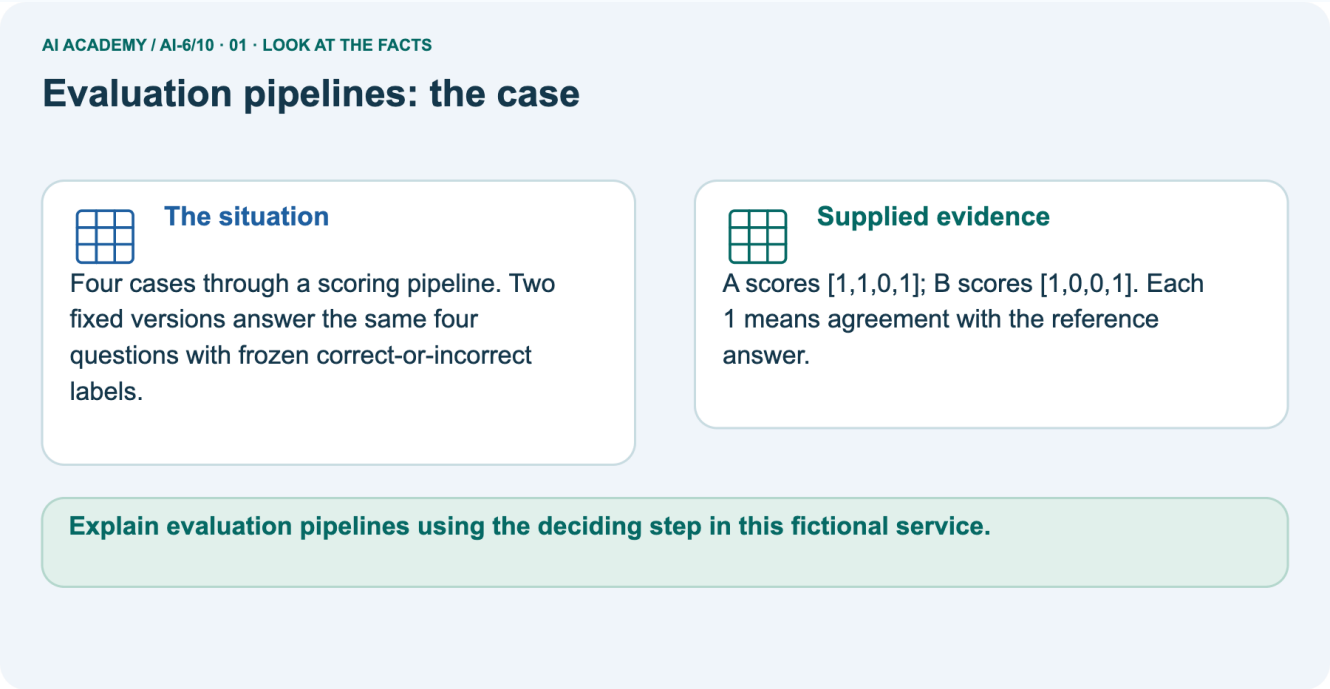
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Evaluation pipelines: the case



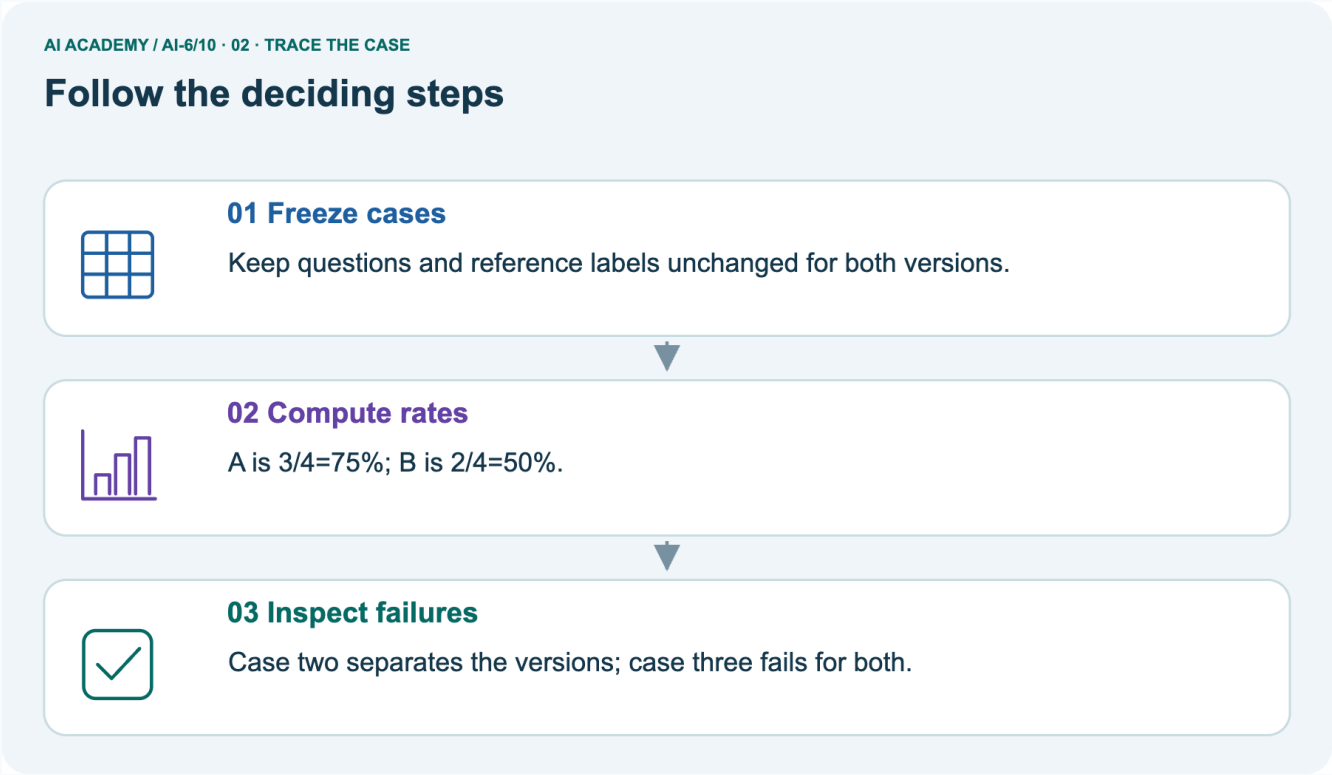
Follow the labelled picture with the matching explanation. Open the vector diagram

Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels. Supplied facts: A scores [1,1,0,1]; B scores [1,0,0,1].

Each 1 means agreement with the reference answer.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

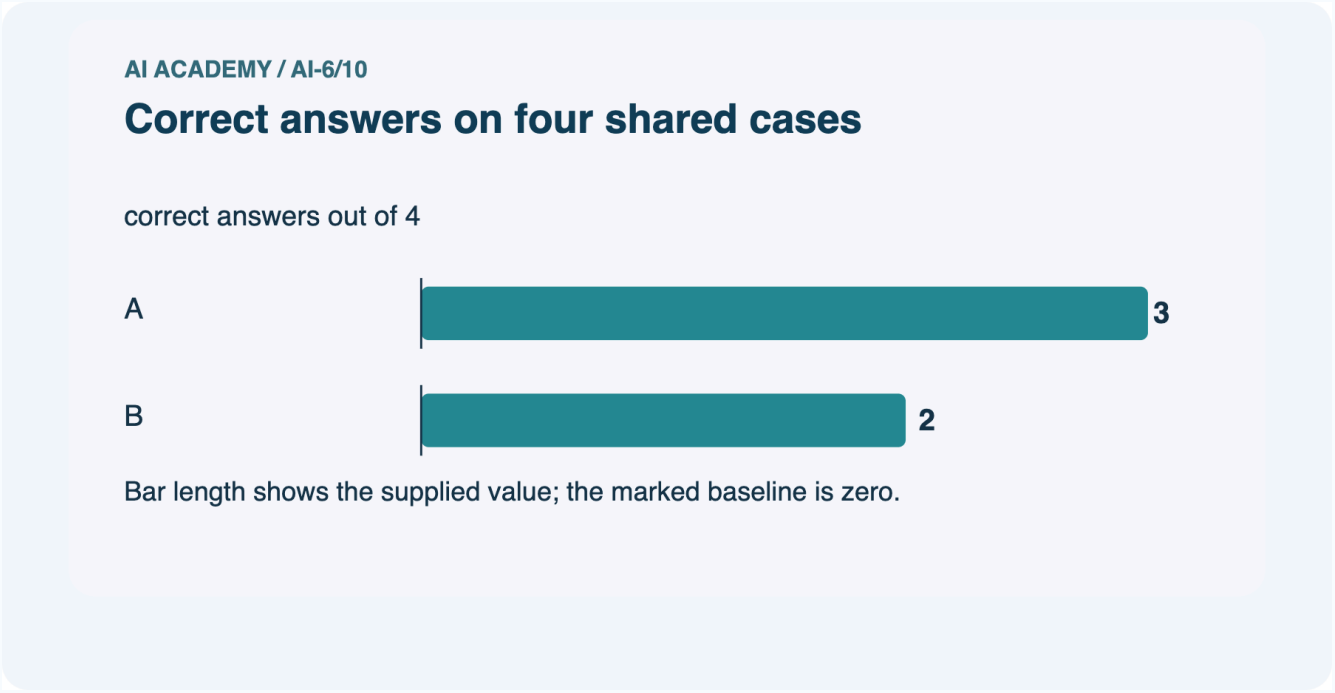


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze cases: Keep questions and reference labels unchanged for both versions. → Compute rates: A is 3/4=75%; B is 2/4=50%. → Inspect failures: Case two separates the versions; case three fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

03. Correct answers on four shared cases

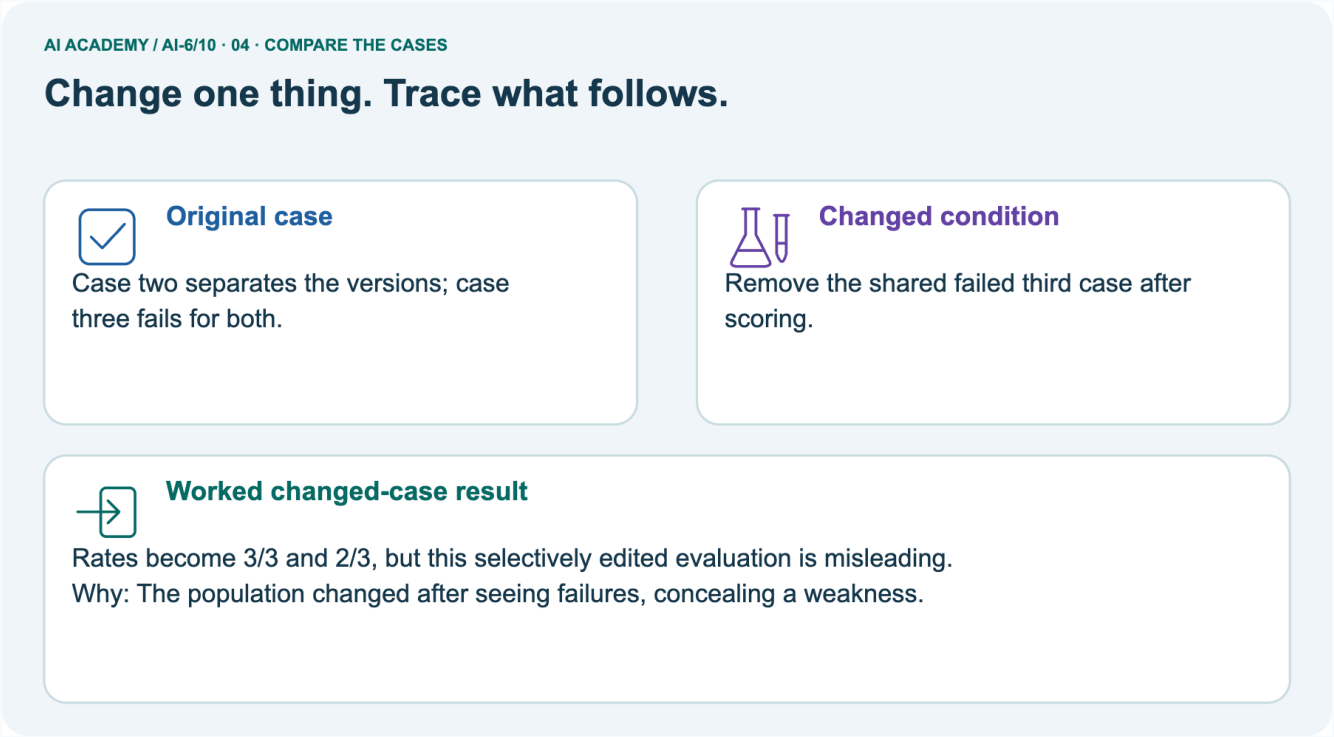


Follow the labelled picture with the matching explanation. Open the vector diagram

Correct answers on four shared cases. Units: correct answers out of 4. A: 3. B: 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



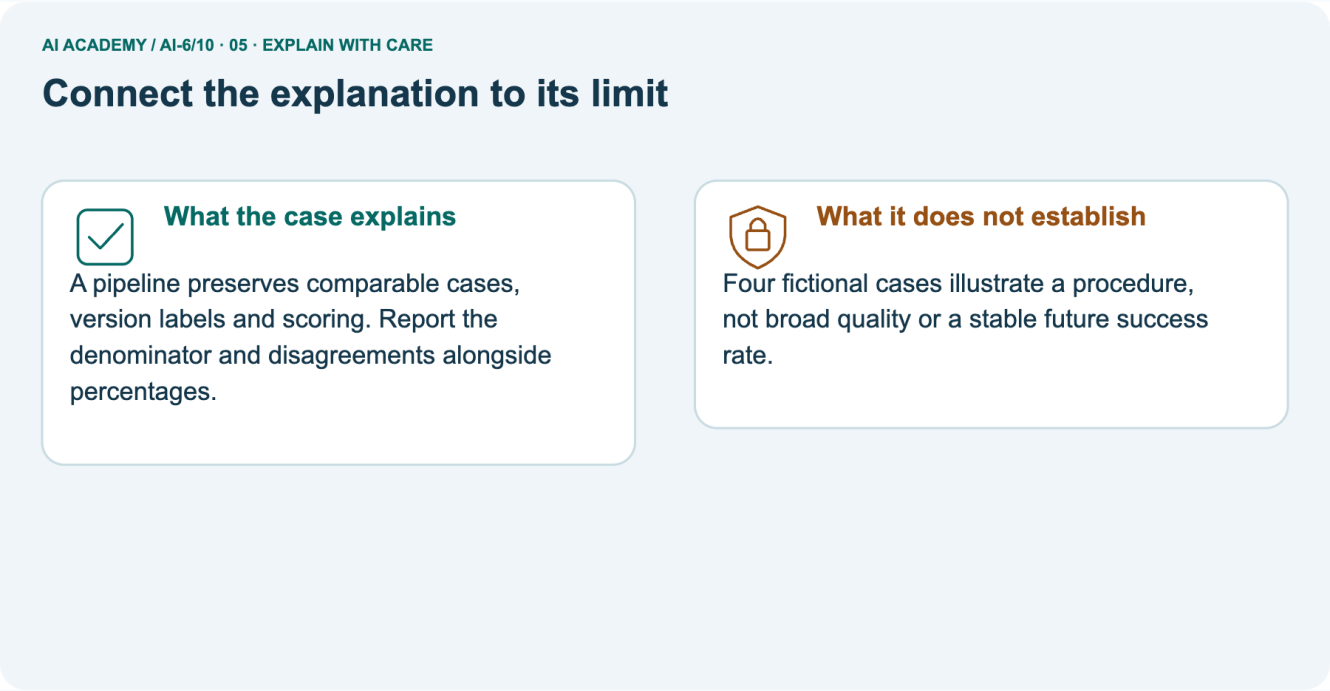
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Case two separates the versions; case three fails for both. Changed condition: Remove the shared failed third case after scoring.

Expected result: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. Reason: The population changed after seeing failures, concealing a weakness.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



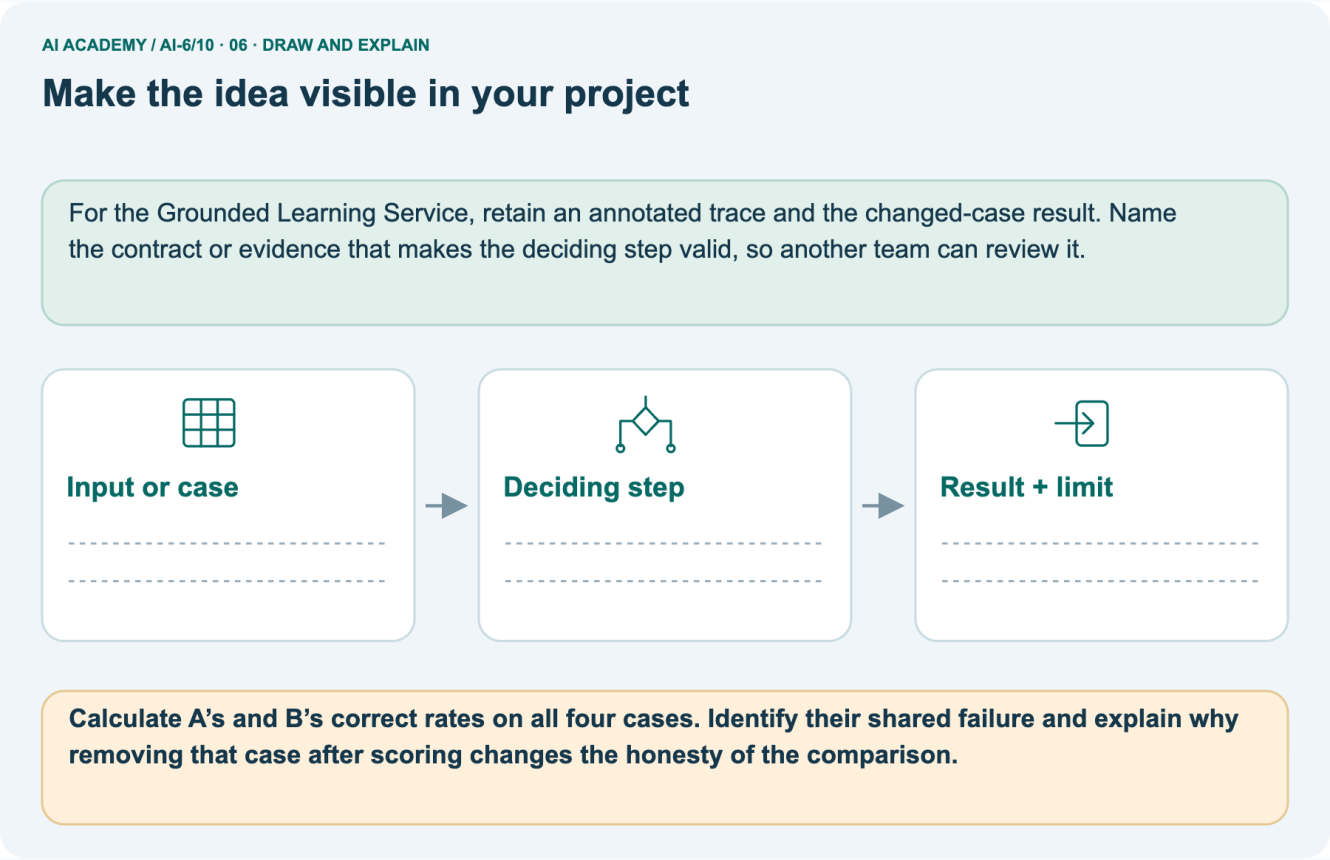
Follow the labelled picture with the matching explanation. Open the vector diagram

A pipeline preserves comparable cases, version labels and scoring. Report the denominator and disagreements alongside percentages.

Limit: Four fictional cases illustrate a procedure, not broad quality or a stable future success rate.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Evaluation pipelines: the case

02. Follow the deciding steps

03. Correct answers on four shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.